Legalyze: An Open-Source Structure-Aware Agentic RAG Platform for High-Precision Legal Intelligence

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Abstract

Artificial Intelligence holds transformative potential for judicial systems, yet the deployment of Large Language Models (LLMs) in statutory law remains severely restricted by hallucinations and unreliable legal citations. In this paper, we present **Legalyze**, an open-source, structure-aware agentic Retrieval-Augmented Generation (RAG) platform tailored specifically for Pakistani jurisprudence. Legalyze addresses structural loss in standard chunking through **Structure-Aware Hierarchical Indexing (SAHI)**, which preserves constitutional, criminal, and procedural statutory boundaries alongside a dual-stream hybrid retriever (BM25 + Dense Semantic Embeddings) fused via Reciprocal Rank Fusion (RRF). A five-tier multi-agent reasoning architecture coordinates specialized statutory retrieval, cross-statute consistency verification, and evidence grounding.

We evaluate Legalyze using a reviewer-blinded protocol with three independent Advocates of the High Court of Pakistan across 67 complex legal queries spanning a 1.2-million-token multi-statutory corpus. Legalyze achieves a **5.5% pooled hallucination rate** (4.5% consensus-adjudicated), representing a **90.0% relative reduction** over Zero-Shot GPT-4o (54.7%) and a **76.0% reduction** over Standard Flat RAG (22.9%). All pairwise performance gains are confirmed statistically significant by paired inferential testing (Cochran's $Q = 37.57, p = 6.93 \times 10^{-9}$; GEE odds ratio = 20.88× over Zero-Shot). Legalyze provides a reproducible, domain-adapted blueprint for high-precision legal intelligence under human expert oversight.

Keywords: Legal AI, Retrieval-Augmented Generation (RAG), Agentic AI, Hallucination Reduction, Hierarchical Indexing, Pakistani Jurisprudence, Multi-Agent Systems.

1. Introduction

Large Language Models (LLMs) have demonstrated substantial capabilities in legal text summarization and drafting, yet their application to specialized statutory jurisdictions remains impeded by legal hallucinations—generating plausible-sounding but legally inaccurate or fabricated citations [**?**, **?**]. In common-law and hybrid legal jurisdictions such as Pakistan, legal reasoning depends strictly upon hierarchical statutory hierarchies (Articles, Sections, Schedules) and binding superior court precedents.

Standard Retrieval-Augmented Generation (RAG) architectures segment legal enactments into arbitrary, fixed-length token chunks (e.g., 500 tokens). This flat segmentation destroys hierarchical boundaries, frequently separating operative legal conditions from exceptions, penalties, and appellate jurisdictional thresholds [**?**]. Consequently, standard RAG frequently retrieves disconnected fragments, leading to misgrounded legal reasoning.

To overcome these structural limitations, we introduce **Legalyze**, an open-source, structure-aware agentic RAG platform engineered specifically for Pakistan's legal corpus. Legalyze combines:

1. **SAHI Parser:** A Structure-Aware Hierarchical Indexing parser that models legal statutes as semantic tree structures.
2. **Dual-Stream Hybrid Retrieval:** Combining keyword-exact BM25 and dense vector semantic search using Reciprocal Rank Fusion (RRF).
3. **Multi-Agent Reasoning Architecture:** Coordinating retrieval, statutory synthesis, and strict grounding verification agents.

2. Key Empirical Findings

Under a blinded evaluation protocol with three active High Court Advocates across 67 legal queries spanning 1.2 million tokens:

- **Hallucination Elimination:** Reduced the pooled hallucination rate from 54.7% (Zero-Shot) and 22.9% (Standard RAG) down to **5.5%** (consensus: **4.5%**), achieving a **90.0% relative reduction**.
- **Omnibus Significance:** Cochran's $Q = 37.57$ ($df = 2, p = 6.93 \times 10^{-9}$), confirming systemic across-architecture improvements.
- **GEE Clustered Regression:** Standard RAG exhibits 5.13× higher odds of hallucination, and Zero-Shot exhibits 20.88× higher odds compared to Legalyze ($p = 1.52 \times 10^{-8}$).
- **Inter-Rater Reliability:** Fleiss' $\kappa = 0.743$ across 603 blinded reviewer assessments.

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